

# TrES-5b

R-1974

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-5\_150614 · created 2026-08-07T00:11:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457187.8545 +/- 0.0018 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.149 +/- 0.022                 |
| Transit depth (Rp/Rs)^2                           | 2.23 +/- 0.65 [%]               |
| Orbital Inclination (inc)                         | 83.72 +/- 1.96                  |
| Airmass coefficient 1 (a1)                        | 0.973 +/- 0.016                 |
| Airmass coefficient 2 (a2)                        | 0.025 +/- 0.025                 |
| Scatter in the residuals of the lightcurve fit is | 1.7 %                           |
| Best Comparison Star                              | #1 - [261, 321]                 |
| Optimal Method                                    | PSF photometry                  |
| Transit Duration (day)                            | 0.069 +/- 0.015                 |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R* fit vs published    | 0.149 ± 0.022 vs 0.142 (deviation 0.32σ) |
| Duration fit vs published | 0.069 d vs 0.0758 d (deviation 0.46σ)    |
| Mid-time O–C              | -4.4 min                                 |
| Depth SNR                 | 6.8                                      |
| Residual scatter          | 1.7 %                                    |

## Light curve

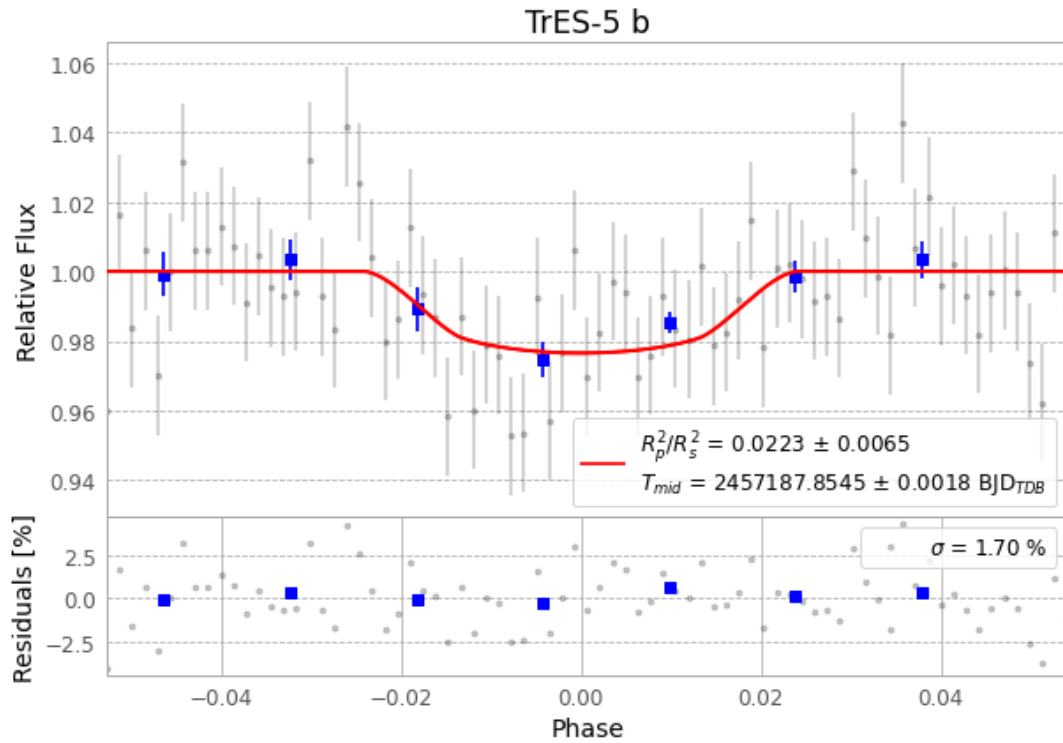


Figure 1 — FinalLightCurve\_TrES-5 b\_2015-06-13.png (SHA-256 d30e2618646efb84...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [126, 241], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 67c5cd624a64acd2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit fd74a91

## Data provenance

77 FITS frames (51 MB), TRES-5150614063612.FITS → TRES-5150614102412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-150614.log (2abcaa2e9f2d...), FinalLightCurve\_TrES-5 b\_2015-06-13.png (d30e2618646e...), FinalLightCurve\_TrES-5 b\_2015-06-13.csv (b2c1e037ddbd...)

## Compute resources

Wall time 122.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 00:11Z.